

Makhura Kamogelo

Week 8 practical

Tasks:

Part A: NAT Overload (PAT)

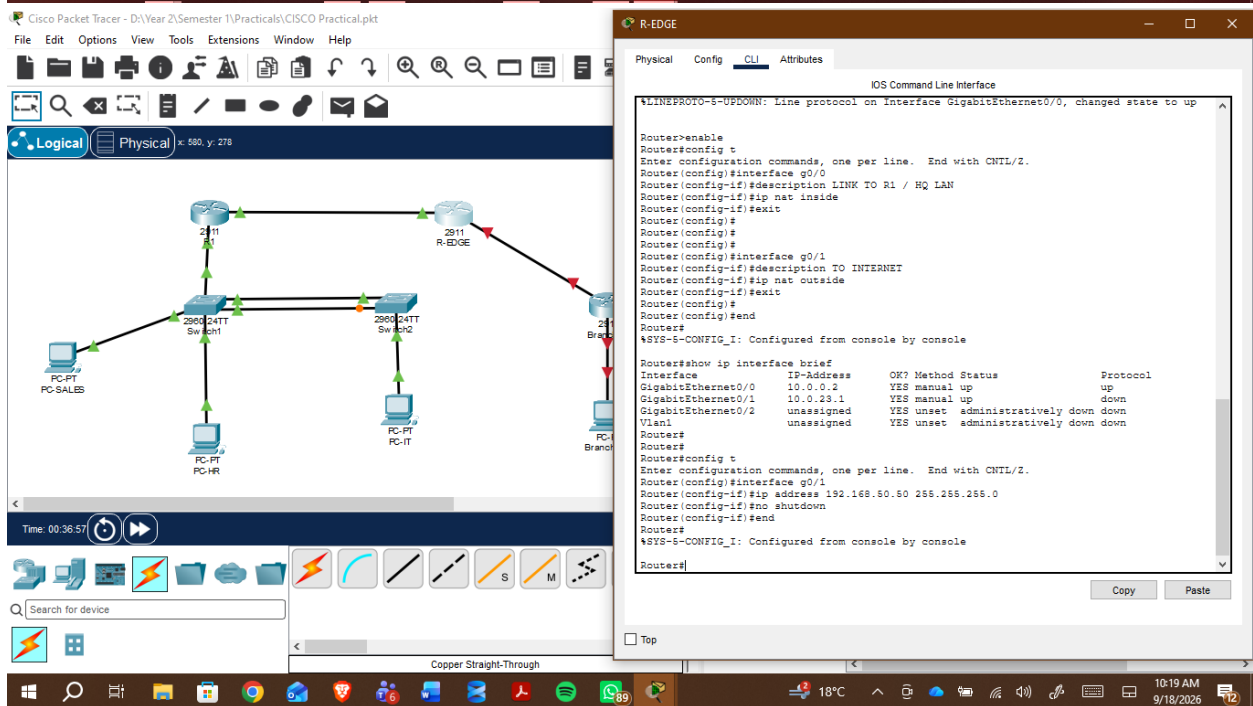
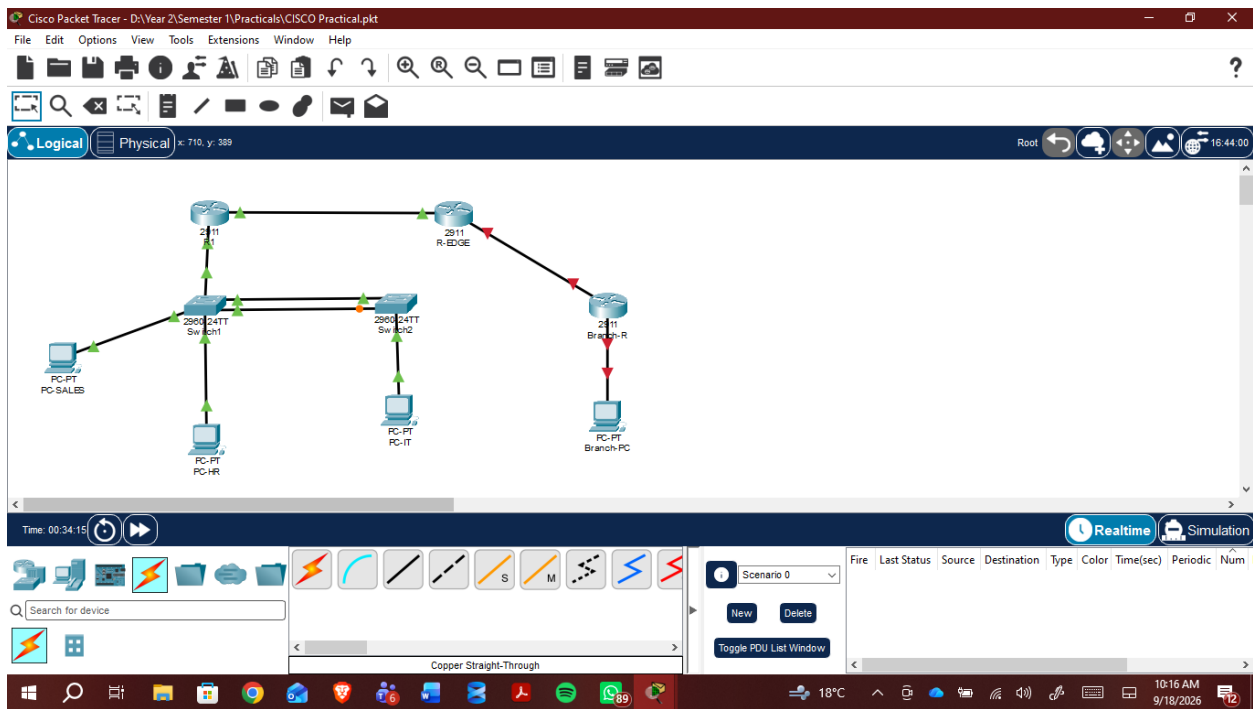
1. interface g0/0 ip nat inside
 2. interface g0/1 ip nat outside
 3. access-list 1 permit 192.168.0.0 0.0.255.255
 4. ip nat inside source list 1 interface g0/1 overload
-
1. show ip nat translations / show ip nat statistics

Part B: GRE Tunnel to the Branch

1. On R-EDGE: interface tunnel 0 ip address 172.16.1.1 255.255.255.252 tunnel source g0/1 tunnel destination [BRANCH-R's public IP]
2. On BRANCH-R: interface tunnel 0 ip address 172.16.1.2 255.255.255.252 tunnel source g0/1 tunnel destination [R-EDGE's public IP]
3. Add a route so branch traffic uses the tunnel, then verify: show interface tunnel 0 ping [branch LAN from HQ]

Part C: Secure the Tunnel with a Site-to-Site VPN

1. Configure an IPsec site-to-site VPN (ISAKMP/IPsec policy, transform set, crypto map) between R-EDGE and BRANCH-R protecting the GRE tunnel traffic.
2. Verify the VPN/security association is established.



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File Edit Options View Tools Extensions Window Help

Logical Physical x: 551, y: 151

Time: 00:38:26

Search for device

Copper Straight-Through

R-EDGE

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface g0/1
Router(config-if)#description TO INTERNET
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.0.0.2        YES manual up          up
GigabitEthernet0/1  10.0.23.1       YES manual up          down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down

Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.50.50 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 004, y: 305

Time: 00:39:21

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Copper Straight-Through

R-EDGE

Physical Config CLI Attributes

IOS Command Line Interface

```
GigabitEthernet0/1  10.0.23.1       YES manual up          down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down

Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.50.50 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

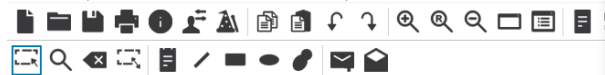
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show access-lists 1
Standard IP access list 1
    permit 192.168.0.0 0.0.255.255

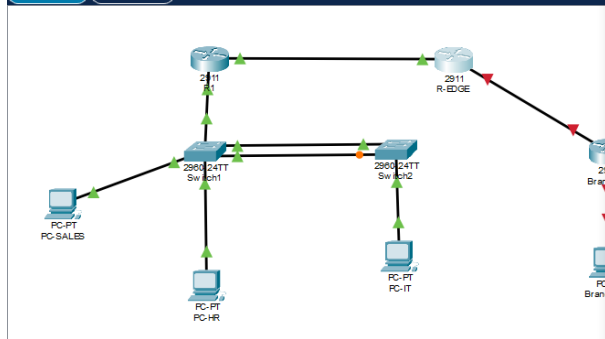
Router#
```

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Logical Physical x: 550, y: 208



Time: 00:41:15



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Copper Straight-Through

IOS Command Line Interface

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

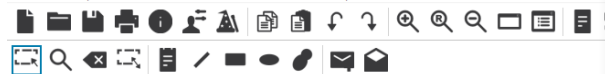
Router#write memory
Building configuration...
[OK]
Router# show ip nat translations
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show access-lists 1
Standard IP access list 1
permit 192.168.0.0 0.0.255.255

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface tunnel 0
Router(config-if)#
%LINK-6-CHANGED: Interface Tunnel0, changed state to up

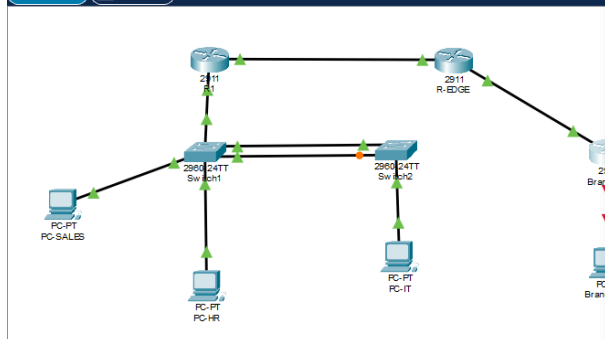
Router(config-if)#ip address 172.16.1.1 255.255.255.252
Router(config-if)#tunnel source g0/1
Router(config-if)#tunnel destination 198.51.100.1
Router(config-if)#tunnel mode gre ip
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router#
```

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Logical Physical x: 559, y: 220



Time: 00:43:29



Search for device



Copper Straight-Through

IOS Command Line Interface

```
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface tunnel 0
Router(config-if)#
%LINK-6-CHANGED: Interface Tunnel0, changed state to up

Router(config-if)#ip address 172.16.1.2 255.255.255.252
Router(config-if)#tunnel source g0/1
Router(config-if)#tunnel destination 203.0.113.1
Router(config-if)#tunnel mode gre ip
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.100.100 255.255.255.0
Router(config-if)#no shutdown

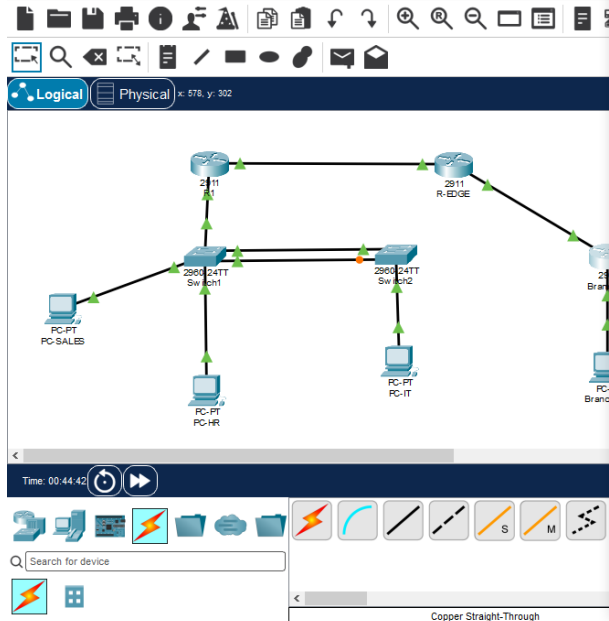
Router(config-if)#
%LINK-6-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

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Branch-R

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#tunnel destination 192.0.113.1
Router(config-if)#tunnel mode gre ip
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.100.100 255.255.255.0
Router(config-if)#no shutdown

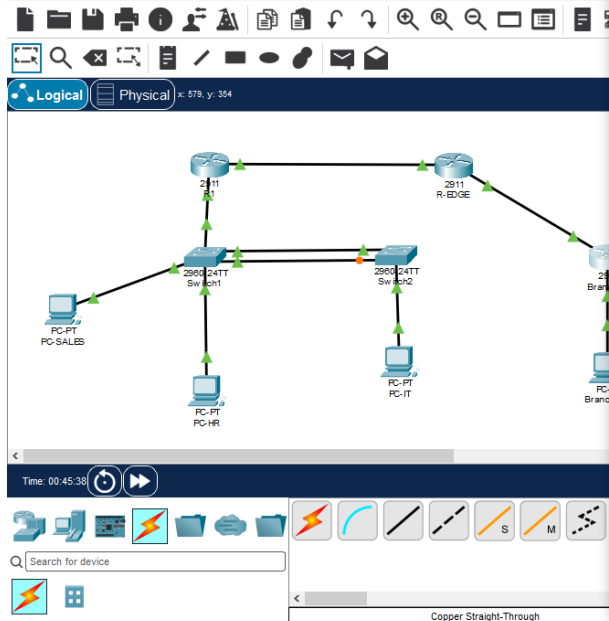
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
% Invalid input detected at '^' marker.

Router(config)#interface g0/1
Router(config-if)#ip address 192.168.160.160 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
```

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R-EDGE

Physical Config CLI Attributes

IOS Command Line Interface

```
$SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip nat translations
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show access-lists 1
Standard IP access list 1
 permit 192.168.0.0 0.0.255.255

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface tunnel 0

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up
Router(config-if)#ip address 172.16.1.1 255.255.255.252
Router(config-if)#tunnel source g0/1
Router(config-if)#tunnel destination 198.51.100.1
Router(config-if)#tunnel mode gre ip
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#ip route 192.168.2.0 255.255.255.0 tunnel 0
% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.2.0 255.255.255.0 172.16.1.2
Router(config)#
Router(config)#
```

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Logical Physical x: 597, y: 349

Time: 00:46:24

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Copper Straight-Through

Branch-R

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#end
Router#
%SYS-S-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
% Invalid input detected at '^' marker.

Router(config)#interface g0/1
Router(config-if)#ip address 192.168.150.150 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#end
Router#
%SYS-S-CONFIG_I: Configured from console by console

Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 tunnel 0
% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.2.0 255.255.255.0 172.16.1.3
Router(config)#
Router#
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 537, y: 224

Time: 00:47:14

Search for device

Copper Straight-Through

R-EDGE

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#
%SYS-S-CONFIG_I: Configured from console by console

Router#show interface tunnel 0
Tunnel0 is up, line protocol is down (disabled)
Hardware is Tunnel
Internet address is 172.16.1.1/30
MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 192.168.50.50 (GigabitEthernet0/1), destination 198.51.100.1
Tunnel protocol/transport GRE/IP
Key disabled, sequencing disabled
Checksumming of packets disabled
Tunnel TTL 255
Fast tunneling enabled
Tunnel transport MTU 1476 bytes
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 input packets with dribble condition detected
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 unknown protocol drops
0 output buffer failures, 0 output buffers swapped out

Router#
Router#
Router#
Router#
```

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Reflection Questions (answer these in your submission):

1. A GRE tunnel on its own isn't encrypted — what does adding IPsec actually protect that GRE alone doesn't?

GRE alone does not encrypt. It only encapsulates private IP packets inside a public GRE header so they can cross the internet. The data is still clear-text and can be sniffed or modified. Adding IPsec adds:

Confidentiality (AES encryption), **Integrity** (SHA HMAC to detect changes), **Authentication** (pre-shared key proves who sent it), and **Anti-Replay** protection. So GRE builds the tunnel, IPsec secures it.